n	10	100	10000	100000	100000000
A_n	2.593742460	2.704813829	2.718145927	2.718268237	2.718281815

We notice that as n gets really large, A_n values seem to be getting closer to 2.718281815..... Actually A_n values approach a real number e, an irrational number. 2.718281815 is an approximation to e. So the compound interest formula becomes $A = Pe^{rt}$, where r is the interest rate and P is the principal and t is the number of years. This is called Continuous Compounding.

Example 2.31 (i) Simplify: $(x^{1/2}y^{-3})^{1/2}$; where $x, y \geq 0$.
(ii) Simplify: $\sqrt{x^2 - 10x + 25}$.

Solution:
(i) Since $x, y \geq 0$, we have $(x^{1/2}y^{-3})^{1/2} = x^{1/4}/y^{3/2}$.
(ii) Observe that $\sqrt{x^2 - 10x + 25} = \sqrt{(x-5)^2} = |x - 5|$.



(i) $(x^{1/4})^4 = x$ but $(y^4)^{1/4} = |y|$.
Observe that $x^{1/4}$ is defined only when x is positive. But y^4 is defined even when $y < 0$.
Now $(y^4)^{1/4}$ is a positive number whose fourth power equals y^4. So it has to be $|y|$.

(ii) $(x^8.y^4)^{1/4} = x^2|y|$.

(iii) Let u, v, b be rational numbers where b is positive.
Let us suppose they are not squares of rational numbers.
Then $u + v\sqrt{b}, u - v\sqrt{b}$ are called **conjugates**.
Observe that $(u + v\sqrt{b})(u - v\sqrt{b}) = u^2 - bv^2$ is now rational.
Thus, if an expression such as $u + v\sqrt{b}$ appears in the denominator we can multiply both the numerator and denominator by its conjugate, namely, $u - v\sqrt{b}$, to get a rational number in the denominator.

(iv) Using $(u\sqrt{a} - v\sqrt{b})(u\sqrt{a} + v\sqrt{b}) = u^2a - v^2b$, it is possible to simplify expressions when $u\sqrt{a} + v\sqrt{b}$ occurs in the denominator.

Example 2.32 Rationalize the denominator of $\dfrac{\sqrt{5}}{(\sqrt{6} + \sqrt{2})}$.

Solution:
Multiplying both numerator and denominator by $(\sqrt{6} - \sqrt{2})$,we get
$$\frac{\sqrt{5}}{(\sqrt{6} + \sqrt{2})} = \frac{\sqrt{5}(\sqrt{6} - \sqrt{2})}{(\sqrt{6} + \sqrt{2})(\sqrt{6} - \sqrt{2})} = \frac{(\sqrt{30} - \sqrt{10})}{4}.$$

Example 2.33 Find the square root of $7 - 4\sqrt{3}$.

Solution:

Let $\sqrt{7 - 4\sqrt{3}} = a + b\sqrt{3}$ where a, b are rationals.
Squaring on both sides, we get $7 - 4\sqrt{3} = a^2 + 3b^2 + 2ab\sqrt{3}$. So, $a^2 + 3b^2 = 7$ and $2ab = -4$.